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The 7 pillars of TRIZ philosophies

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ABSTRACT

TRIZ, the theory of inventive problem solving, has known as one of the most effective tools for conceiving engineering designs and solving difficult problems. The importance of TRIZ has been acknowledged in various fields where its applications inspire great results. Most researchers focus on the practical usage of the tools or development of the tools without knowing the fundamental philosophies that support TRIZ. Classical TRIZ has only 4 philosophies as pillars of TRIZ. The aim of this paper is to present an augmented version of TRIZ philosophies, and bridge the gap between the guiding fundamental philosophies and effective practical usage. Further, some important directional research opportunities of TRIZ methodology are suggested for future development.

1. Background

Among all Innovation Methods (IM), Systematic Innovation (SI) is a field which concerns about developing or using systematic methods/ processes to generate innovative ideas for technical, strategic, or business aspects of opportunity identification and/or problem solving (Sheu & Lee, 2010; Sheu, 2016**)**. The concepts of SI are from studies of prior humans' innovative ideas/wisdom, which is known as Human originated SI or natural inspired phenomena such as biomimicry. The systematic innovation method utilizes process and techniques that can help to solve the problems. The working principles of these innovative approaches were extracted and then re-used to generate innovative ideas. Fig. 1 presents a proposed classification of innovation methods where systematic innovation plays a critical role (Sheu, 2010; Govindarajan, Sheu, & Mann, 2019). Fig. 2 explains the difference between TRIZ and other systematic innovation tools. Based on authors observations, even though some 200 others tools can be classified as systematic innovation (SI) tools, such as de Bono' lateral thinking, 6 thinking heads, morphological matrix, etc. (Silverstein et al., 2012), most of other SI tools are individually designed to solve specific category of problems and are not synergized with one another. Unlike most other SI tools, TRIZ tools can be used to solve wide ranges of problems such as engineering problems, management problems, identifying innovative products and services, patent circumvention/development/ and deployments, identifying strategies and business model elements, and working together synergetically with other tools such as value engineering, 6 sigma, lean tools, etc.

1.1. Introduction

TRIZ, the Russian acronym for "Theory of Inventive Problem Solving" (Rantanen & Domb, 2007), is the quintessence of a thorough study of about 40,000 technology patents by Genrich Altshuller and his colleagues. Altshuller summarized and draw out certain regularities and basic patterns which governed the processes of solving problems, which can aid in creating new ideas and innovation. TRIZ can analyze problems from many viewpoints such that user can break the barrier of psychological inertia and leverage prior wisdom to generate ideas for problem-solving. Since its introduction outside of the soviet world, in the west particularly, TRIZ has proved itself as a powerful problem-solving tool as a significant innovation method. Many researchers have validated effectiveness of TRIZ (Okudan et al., 2012; Sheu & Hou, 2013; Sheu & Chiu, 2017; Wang, 2017; Sheu, Hong, & Ho, 2017; Lee, Zhao, & Lee, 2019). Numerous descriptions convey the idea that TRIZ extends a set of principles to potential suggestions. TRIZ is a knowledge-based systematic methodology of inventive problem solving (Savranksy, 2000). Actually, TRIZ is more than a set of principles that describe how technologies and systems evolve. Fey and Rivin (2005) described TRIZ as a methodology for the effective development of new [technical] systems. Livotov (2008) regarded TRIZ as the most comprehensive, systematically organized tool for invention and creative thinking methodology.

TRIZ claims that technology evolution and the way to invention is not a random process, but is predictable and can be governed by certain laws which has analytical logic and a systematic way of thinking that

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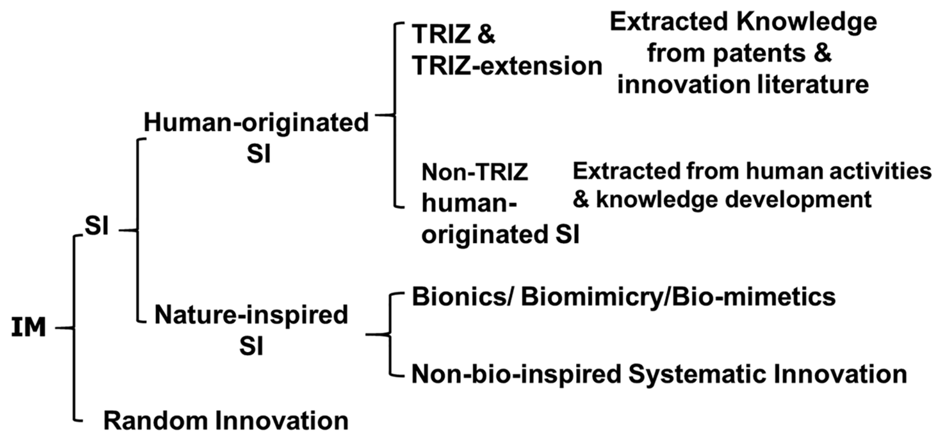

Fig. 1. Classification of Innovation Method (Sheu, 2010; Govindarajan et al., 2019).

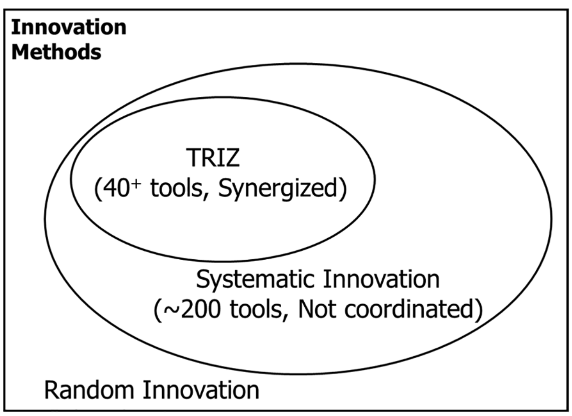

Fig. 2. Relationships among TRIZ/Systematic Innovation/Innovation Methods.

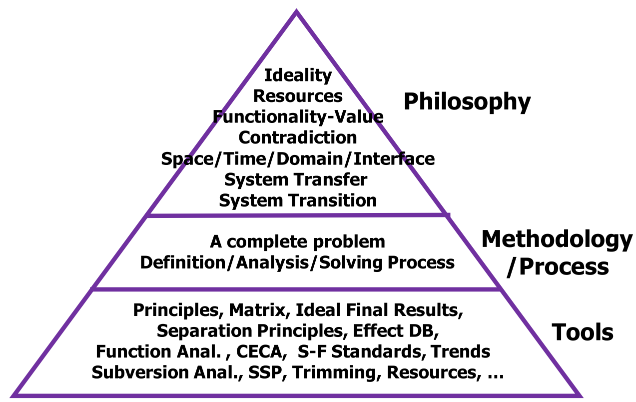

Fig. 3. Hierarchical view of TRIZ (Sheu, 2016 modified from Mann, 2007; Govindarajan et al., 2019).

can provide an overall structure for the application according to the collection of TRIZ tools and techniques (Salamatov and Souchkov, 1999; Savranksy, 2000). Compared to other problem-solving methods such as mind mapping, brainstorming, and lateral thinking, TRIZ helps identify problems and offers possible solutions to users. Although TRIZ has been described differently by researchers as a methodology, a toolkit, a science, a philosophy (Nakagawa, 2001), etc., it is capable of finding solutions to technical problems and yielding innovation in technical systems, is unanimously acknowledged.

Furthermore, TRIZ has been proved to be applicable on wide range of problems. The applications of TRIZ are numerous. A study by Poppe (2002) outlined the possible ways of introducing TRIZ into process industry. The examples available in the open literature describe the use of TRIZ to product design and improvements, equipment design and improvements (Busov, Mann, & Jirman, 1999), process design and improvements, trends identification, and technology forecasting, etc.

While scholars agree on TRIZ as an effective tools set for innovation and problem solving based on the identification of physical or technical contradictions and endorse its usefulness, not very much attention was given to the in-depth understanding of the fundamental philosophies that effect innovative solutions.

This paper investigates seven pillars of TRIZ philosophy to reveal insights about how the philosophy and values behind TRIZ and the main reason why it can work so well and address so many different categories of problems. In this study, we also point out how the TRIZ tools work and their relationships with the TRIZ philosophies.

1.2. Hierarchical view of TRIZ

Fig. 3 presents a hierarchical view of TRIZ (Mann, 2007, Sheu, 2016). At the bottom level of the TRIZ hierarchy is a series of tools and techniques which are used to actually solve practical problems – especially engineering problems. On top of the practical tools, there is a methodology or process that links the various tools to solve problems with synergy and in proper sequence. These tools or methodology, are based on some fundamental philosophies, which are considered as pillars of TRIZ. The authors believe that because of these fundamental thinking philosophies, TRIZ can be used to handle a wide range of problems innovatively such as engineering problems, management problems, etc., as indicated in section 1.

2. TRIZ main philosophies: 7 pillars of TRIZ

The traditional TRIZ concept contains 4 philosophies (Mann, 2000). They are Ideality, Resources, Functionality, and Contradiction. Mann (2007) identified Space/Time/Interface as the fifth pillar of TRIZ. System Transfer and System Transition were respectively proposed as the sixth and seventh pillars of TRIZ by Sheu in class teaching in 2016 but was not published then.

2.1. Ideality

Ideality derives from "the ideal machine", an arbitrary system which has all its parts performing at the greatest possible capacity (Altshuller, 1999). Ideality is a measure of how close a system is to the best it can possibly be, that is, the ideal machine (or the ideal final result (IFR)). Ideality is defined as shown in equation (1). The benefits are the useful functions provided by the system while costs and harms are its unwanted outputs or waste products (also regarded as harmful functions) of the system

The best value of ideality is infinity. When Ideality equals ∞, the Ideal Final Result (IFR) achieved. The ultimate ideal result is to achieve the required functions, and without any cost or harm (cost + harm = 0) as expressed in Equation (1). Even though in reality the infinity level may not be achievable, it provides the direction the system designer ought to strive for. The ideality idea allows us to put the psychological inertia of pre-set current constraints aside and think of ways to achieve the ideality as the goal. If the best ideality can not be achieved, step-back process can be used to find second choice or third choice, etc. The results are often better than those solutions coming from considering constraints first. The solutions may also include self-service principles such as self-cleaning, self-cooling machines, etc.

$$
\begin{aligned}
\text{Ideality} &= \text{Benefits}/(\text{Costs} + \text{Harms}) \\
&= (\text{Perceived Benefits})/(\text{Costs} + \text{Harms}) \\
&= (\text{Useful functions})/(\text{Harmful functions})
\end{aligned}
\tag{1}
$$

2.2. Resources

In the TRIZ environment, a resource can be any substance, field (energy), function, attribute, space, time, information, or even vacuum, void, or "nothing" which can be used toward some purpose (Sheu, 2013). There are two key ideas under this pillar:

1) Use the resources that are not used, discarded, not designed to, or even seeming irrelevant to your purpose, but can be used to achieve your desirable functions (Useless to Useful: U2U). For example, the traffic of Tokyo metro station has more than four million commuters everyday, and the energy wasted by passengers stepping on the floor is not utilized. The subway engineers installed piezoelectric materials on the floor and they were able to convert the unused stepping energy into electricity good enough for use at tickets gates and other signage as shown in Fig. 4.
2) Convert harmful substances into useful resources (Harm-to-Help; H2H): For example, the principle of thermal power generation transforms "thermal energy into electrical energy", through the production of steam from combustion which serve to drive the turbine and rotate the generator. The process of burning coal or oil emits flue gas which contains particulate, sulfur dioxide, nitrogen oxides and carbon dioxide, which will not only cause air pollution, but also endanger human health. Meanwhile, these extremely high temperature gases or dust will also form thermal pollution. In the early years, people usually build a high chimney to let the particle pollution and thermal pollution reach a level of hundreds of meters above ground that would dilute the pollution density so that when the particle pollutants and temperature pollution reach the ground level, they become bearable by humans. Later, new technology takes the flue gas to be discharged to the atmosphere to electrostatic dust collector so that more than 99.9% of the particles are collected and highly pressed into building bricks so that harmful pollutants become useful building bricks. The remaining high-temperature cleaned flue gas is then fed into a heat exchanger to pre-heat the fresh air before it comes to the furnace to burn fuels. As the fresh air carries more energy before it can use less fuel to heat water into steam and produce more electricity. It can be concluded that "the harmful particulate pollution becomes useful building materials", and "the harmful thermal pollution becomes more electrical energy".

Fig. 4. Tokyo piezoelectric subway station. .
Source : https://www.treehugger.com/renewable-energy/japan-producing-electricity-from-train-station-ticket-gates.html

It may be worthwhile to compare the Zero waste of Lean concept with the TRIZ resource concept. Although Lean Concept and TRIZ's Resources concept seem to be of two unrelated historical origins. It is known that the spirit of "zero waste" in Lean concept is to fully utilize resources within or around the operational system in the life cycle of the operations (Sheu, 2010). The main thrust of Lean focuses on the maximum usage of resources that are within or around the system to achieve our goals. By "zero waste" we can make the most of the existing resources we are deploying and maximize their benefits. The authors consider that the TRIZ concept of resource is one or two steps further than Lean concept. First, most people wanting to get resources mainly obtain them from within or around the system and willing to pay for it instead of attempting to get free or near free resources from totally irrelevant sources. Second, most people don't see harmful things as "resources". They usually desire to avoid or eliminate harmful things instead of taking advantage of the harmful things. Breaking these two psychological inertia, TRIZ Resources proposed two step-further concepts than the Lean concept. Even though they are developed individually and separately, the lean concept can be considered as a variant of resource concept.

2.3. Functionality-Value

While most companies consider product is what they are working for. TRIZ's Functionality considers that function is what really counts, not the product itself. Therefore, the mission of R&D team should be achieving functionality while minimizing physical product. This brings a distinct paradigm shift to allow engineers to jump out of company's current product constraints and consider possibly remote or different products to achieve the same functionality. With a functionality-oriented thinking, there will be more opportunities to break out "the product" mindset that will lead to think about different alternatives that can serve the same or better functions. This is level 1 "think outside the box" mentality.

Sheu (2016) extended the Functionality idea to claim that Value is what is really needed in his class teaching. If one can achieve the same or higher level of value, the function of the current level can be eliminated or replaced with other function. The authors further propose this Value-Function-Effect-Component hierarchy. Solving a problem can be at different levels of the hierarchy: 1) Achieving higher level value instead of the current one. 2) Achieving current value with different function. 3) Achieving same function with different action principle (effect). 4) Achieving same action principle with different components. The higher the level is satisfied, the higher level of innovativeness can be achieved. This concept can also be use in systematic patent circumvention, enhancement, and regeneration. Taking washing machine as an example. The main function is to wash clothes, the situation can be represented as following in Table 1.

The Function "clean clothes" has the value "clothes being clean". The following different levels of innovative designs can thus be considered:

1) Satisfy at the Value level instead of functional level. Instead of using a washer to perform the "clean clothes" function, we can use "repel particles" to satisfy the value of "cloth being clean" and never

Table 1
Functional analysis of the washing machine.

Industry	Function Carrier	Function	Object	Value
Washer Industry	Washer	Cleans	Clothes	Clothes being clean

Fig. 5. Ultra ever dry applied to gloves.

obtain dirty with the use of Ultra-ever-dry coating material available at (UltraTech International, Inc., 2012). It is a coating product where Nanotechnology is applied to coat a thin film on its surface. This film repels water, oil, other liquids, and particles as shown in Fig. 5 (Davies, 2013). On the left, the untreated glove becomes dirty while the right treated one remains stain proof due to the application of Ultra-ever-dry.

Note that, it is also possible to achieve a higher level of value and by-pass the current level of value to achieve even higher innovativeness. For example, if the purpose of cloth being clean is to please a new friend, we may prepare a small gift to please the new friend. This satisfaction at higher-level value will open up a paradigm-shift opportunity in a different industry.

2) Satisfy the functionality but replace with different action principles to achieve the same functionality. For example, instead of water physically dissolve dirt to clean clothes in the washer machine, we can use chemical reaction or ultra sound vibration to clean clothes. This approach may produce other innovative alternatives.

3) Adopt different components or systems to achieve the same action principle or function. For example, instead of using water to carry detergent, we may use other liquid, for example alcohol, or steam to carry detergent and obtain the same results. Great majority of people will use this level of replacement to solve problems. However, this is at the lowest level of innovativeness.

The “Value-Function-Effect-Component hierarchy” allows us to systematically and comprehensively resolve problems with various levels of innovativeness. Resolving problems at the value level almost always creates a new technology or industry that may well eliminate the current technology or industry.

2.4. Contradiction

After analyzing more than 40,000 patents, Altshuller realized that contradiction is the fundamental barrier of all technical advancement. Since most of contradictions are not obvious. One of the main approaches of TRIZ for problem resolution is to use contradictions to formulate problems and analyze these contradictions to solve the problem (Dubois, De Guio, & Rasovska, 2011). The ability to identify contradictions that are unknown or non-obvious to others implies the ability to identify opportunities to innovate or resolve those contradictions can lead to innovation.

There are two major types of contradictions: technical contradictions and physical contradictions.

Fig. 6a. A duck (Sheu, 2016).

Technical contradiction arises when an attempt to improve certain attributes or performance of a system leads to the deterioration of another attribute performance of that system. Physical contradiction arises when there are incompatible requirements on a single parameter of the same system. Identifying and solving contradictions constitutes one of the most fundamental TRIZ problem solving approaches.

2.5. Space/Time/Domain/Interface

The Space-Time-Domain-Interface (STDI) is a minor extension from Mann’s Space-Time-Interface (Mann, 2007) to cover more comprehensive situations.

STDI implies that in order to solve a difficult problem, we often need to jump to different spaces, times, domains, and/or interfaces to see the problem better. With the ability to jump to different STDI for different perspectives, we can often understand the problem better which allows us to solve the problem. This is because something difficult to understand seeing from one perspective, but may become easy to understand from a different perspective, as looking at another angle, the same thing can be seen very differently. For example, Fig. 6a presents a duck. But when we turn the duck by 90 degree clockwise it turns into a rabbit as shown in Fig. 6b. These two pictures are exactly the same but being seen differently by turning 90 degree.

Fig. 6b. A Rabbit (Sheu, 2016).

As an example of domain switch, a common usage of the famous Fourier transfer is to convert a function in time domain, *f(t)*, into a corresponding function in frequency domain *g(w)*. A system's behaviors in our human-felt time domain is represented as a function *f(t)*. It may be modeled into a set of differential and/or integral equations which are hard to solve. Applying Fourier transfer, the original time domain equations becomes equations in frequency domain that the differential/integral equations become corresponding arithmetic equations which are in "subtraction" and/or "addition" operations. Solving for arithmetic equations are much easier which we already learned in the primary school. This is a domain change which can not be explained in terms of space, time, or interface. In many duality phenomena, a change in one domain of the duality corresponds to another change in the other domain of the duality. If solving a problem in original domain is difficult, it may be easier to solve the corresponding problem in the other domain. The use of simulation can be considered as using alternative domain to locate and solve the problems in the primary domain. For example, to install a new factory layout usually requires enormous capital investment long time to implement. If the original design is inferior, doing the physical layout directly without simulation may result in a large loss. It is beneficial to debug the designed layout in the domain of virtual software model. Then, observe and fix the problems before actually implementing the physical layout. This is the power of changing perspective to a different domain for problem solving.

People usually have common psychological inertia that lead us to look at situations from a specific angle which we are familiar with. If a problem falls into the categories of problems we have solved before, we often can solve it easily. That's the value of training and experience in action. Common reasons that we are not able to solve a difficult problem are 1) either we have not seen the problem before and our background do not provide us sufficient understanding of the problem, or 2) even though we have seen the problem before, it is presented in a way that we have never seen before. Therefore, we are not able to recognize it, not to mention to solve it. Often times, people may even distort the data to fit a model which we are familiar with without self-awareness. That is why the French writer Anis Nin once said, "We don't see things as they are, we see things as we are". This is the famous "psychological inertia" in action.

The power of STDI allows us to break our psychological inertia to see the problem better thus enabling us to understand and solve the problem. A number of TRIZ tools have the essence of STDI in action, for example, in the common TRIZ problem solving process, the first step of "converting Specific Problem into Model of Problem" itself can be considered as seeing the problem instead of at its original form (Specific Problem) but at an abstracted form (Model of Problem) of the problem, which is change of perspective from different STDI. As such, the STDI concept has quietly embedded in the many TRIZ tools such as contradiction matrix (seeing problem from parameters perspective), su-field analysis (seeing problem from substance-field relationship perspective), functional analysis (seeing problem from function-component relationship perspective), etc.

2.6. System transfer (Tf)

When a problem occurs within a system, the issue of solving the problem can be transferred from within the system to an unrelated system. The novelty of the System Transfer resides in transferring the issue of the current system to a seemingly unrelated system to solve the problem more elegantly and innovatively. Taking the threading needle as an example. When threading, the eye of the needle has to be large for easy putting the thread through the needle eye. However, the needle hole diameter ought to be small in order not to damage the clothes. For one of the system transfer solutions, we can make the hole aperture small, as shown in Fig. 7, to meet the requirements of not destroying the cloth tissue. Then, transfer the requirement of the large pinhole for easy threading to a newly invented threading device as shown in Fig. 8. The hole of the threading device is large when the force is not applied, and smaller enough to get into the pinhole when the lateral compression force is applied.

Fig. 7. Needle with physical contradictions (Sheu, 2016).

Fig. 8. Treading device solution (Sheu, 2016).

The idea of system transfer can also be used in management or strategic area with powerful results. As an illustration, the famous ancient Chinese military story of "Besiege Wei to Rescue Zhao" is given here. In 354 BCE, the state of Wei attacked the state of Zhao and laid siege to its capital Handan. Zhao turned to the neighboring state, Qi, for help. The general and all soldiers of Qi are expecting to go to Handan to the rescue. But the Qi's military advisor, Sun Bin, thinks that heavy casualties will occur even if they win over the strong enemy. It would be unwise to meet the army of Wei head on. So he instead, attacked Wei's capital at Daliang where the city is weak as all strong troops have gone attacking Zhao. When the Wei general Pang Juan heard that the capital was being attacked, he rushed his army back to defend the capital. The army of Wei retreated in haste, and the tired troops were ambushed and defeated at the vulnerable passage of Guiling. Zhao was thus rescued while most of Pang Juan soldiers are killed with no loss of Qi's soldiers at all. This ingenious "System Transfer" can only be conceived by the military genius Sun Bin. However, it is possible to reason to the tactics by the systematic thinking process of system transfer.

2.7. *System Transition* (Ts)

System Transition implies a radical change in the problematic system to solve the problem. System transition involves a paradigm shift in problem solving. Some examples are the phase transition (solid–liquid-gas), the butterfly metamorphosis as shown in Fig. 9 and so on. In management area, cooperate merging, organizational structural change, drastic change of business model, or business process reengineering, are some examples of system transition.

Fig. 9. Butterfly transformation as an example of System Transition (Sheu, 2016).

2.8. Comparison of "Space-time-domain-interface", "System Transfer", and "System Transition

STDI (space/time/domain/interface) implies that we are able to look at the problem from multiple perspectives instead of just the initial viewpoint - the viewpoint we normally see based on our background. STDI has two advantages: 1) We will be able to see the same thing differently compared to seeing it from the initial viewpoint. At certain perspectives, we can see the problem better and understand it better. When we understand a problem better, we are more able to solve it. 2) We will be able to integrate multiple views to get a full picture of the problem conducive for better problem solving. STDI philosophy allows us to understand the problem better which may enable us to solve it. However, STDI does not provide problem solving approach directly. System Transition is an innovative problem solving approach which radically changes the problematic system to solve the problem. It often represents or induces paradigm shifts in problem solving. System Transfer is an unexpected innovative problem-solving approach. Instead of solving the problem itself, which the great majority of problem solver will do, system transfer diverts the attention on the problematic system to another seemly unrelated system and dealing with that system or using that resource to solve the problem – often more elegantly and effectively. The comparison of these three philosophies are depicted in Table 2.

2.9. A^+ TRIZ

Traditional TRIZ has been mostly utilized in analyzing and solving engineering problems which include new and existing product developments/improvements, new and existing process/equipment developments/improvements as indicated in the first category of Table 3. The range of applications of TRIZ has been greatly expanded by various recent researchers and provided many versions of enhanced TRIZ as modern TRIZ and named with various labels such as $TRIZ^+$, GEN-TRIZ, CTRIZ, XTRIZ, TRIZ++, etc. (Sheu, 2016) presented an augmented version of modern TRIZ, named A^+ TRIZ, as shown in Table 2 where the applications of A^+ TRIZ philosophies has been proven effective. More than 20 new tools for use in various applications beyond traditional TRIZ have been developed. In addition, relevant forms are created to facilitate the process of using the tools. It is noted that TRIZ is also very effective in working with various other tools such as Value Engineering (VE), Quality Function Deployment (QFD), Failure Mode and Effect Analysis (FMEA), 6 Sigma, Lean, Kepner-Trego, and Theory of Constraint (TOC), et at, as indicated in Table 2 (Kim, 2018; Wang, 2017). Even though being considered as a less travelled topic, some recent work of TRIZ for software innovation were reviewed by (Govindarajan et al., 2019).

Table 3
Extended Application Areas of TRIZ (SI) (Sheu, 2016).

- ■ **Solving Engineering problems**
- ● New/existing product Development/Improvements
- ● New/existing Process/Equipment Development /Improvements
- ● Patent Circumvention/Enhancements/Regeneration
- ■ **Identifying Innovative Products & Services**
- ■ **Management/Service Applications**
- ● Identify Organizational conflicts & solve them
- ● Establish Innovation Strategies/Business Model
- ■ **Combining with other tools to solve problems**

VE; QFD; FMEA; 6-Sigma tools, Lean, Kepner-Trego, TOC,

3. The link between the guiding philosophies and practical tools

3.1. Existing links between the fundamental philosophies and practical usage

Some existing tools based on these philosophies can be identified as in Fig. 10 and explained below.

In the Fig. 10, the upper ellipses represent the 7 philosophies. The lower boxes contains names of various tools which are guided or supported by corresponding philosophies.

- Ideality itself provides the measures to calculate the goodness of any product. The Ideal Final Result process from (Mann, 2007) is based on ideality concept starting from the highest ideality.
- The method of Demand-Supply Thought Provoking Question (DS-TPQ) in (Sheu & Hong, 2017) was developed from the Resources philosophy to identify unexpected resources. The methods to utilize harmful resources as proposed in (Sheu & Yen, 2020) is also based on the Resources philosophy.
- Function-value philosophy has been used to develop systematic patent regeneration methods in (Sheu, 2019) book.
- Contradiction philosophy is the base for contradiction matrix and separation principles.
- The smart little people and STIC (Space, Time, Interface, Cost) tools are realizations of the STDI philosophy. Effect database (OxfordCreativity, 2019) can also be regarded as looking from function/parameter perspective (domain) to collect all effects that may provide the desirable function/parameter changes. The multi-

Table 2
The comparison of STDI, T_f, and T_s.

Philosophy	The way to work	How to solve the problem	Results
STDI	Provide various viewpoints but not the approach to solve the problem	Do not provide solution. After understanding the problem, the solvers figure out the solutions by themselves.	The solver gains insight & comprehensive understanding to the system and is able to solve it.
System Transfer (T_f)	Provide a direction on how to solve the problem	Dealing with another entity and/or using seemingly unrelated resources, to solve the problem.	Problem is solved elsewhere and often unexpectedly / more elegantly.
System Transition(T_s)	Provide a direction on how to solve the problem	Dealing with the problematic entity directly to solve the problem but using a fundamentally different action principle.	Often change the system drastically and with significant results.

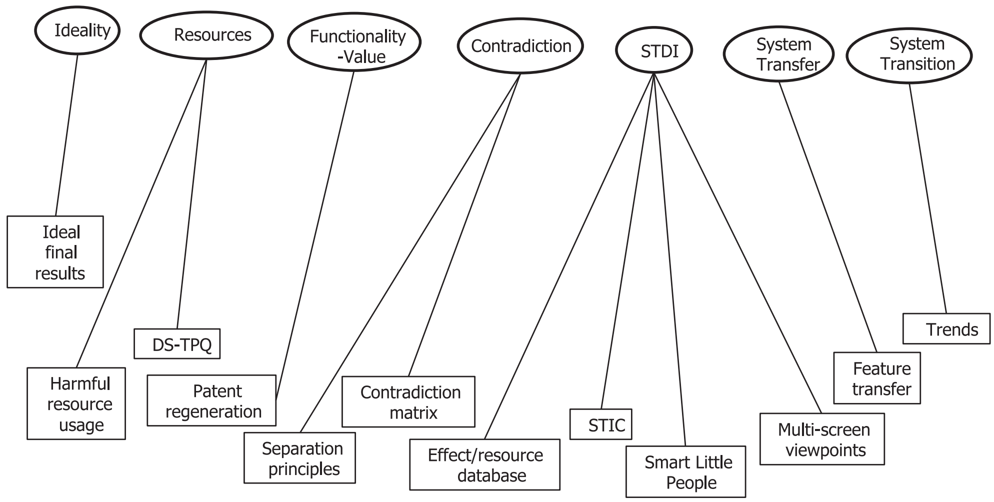

Fig. 10. Links between philosophies and corresponding practical tools.

screen windows tool is looking issues/problems from combinations of different system levels (super-system, system, sub-system) and different times (past-present-future). Therefore, it is a realization of STDI philosophy.
- The Feature transfer tool allows user to transfer features from one system to another to solve problems. As such, it is the philosophy of system transfer in action.
- The trends of engineering systems provide us a way to jump from current technology to a more advanced technology. For example, jumping from cathode ray tube technology to Liquid crystal display technology for display products. That is one realization of System transition concept.

Based on those links, the applications of these philosophies are implicitly realized by the corresponding tools. Based on the above analysis, it is clear that many TRIZ tools do have these philosophies as fundamental principles even though some of them may be implicit and require deep thinking. Many industrial cases of using TRIZ practical tools listed in Fig. 10 are widely available in literature. They can be considered as examples of the usage of the guiding principles. The use of these philosophies are embedded in the applications of their guided tools. This is the nature of being philosophy.

3.2. More opportunities to bridge between the fundamental philosophies and practical usage

Even with the fundamental philosophies behind TRIZ working principles identified and some tools are identified as realization of the philosophies as indicated above, more practical tools are still needed to take us from the current problem situation to achieve these fundamental philosophies systematically. It is clear that more systematic practical tools ought to be developed to fully take advantage of the philosophies for real-world problem solving and opportunity identification in all aspects of applications including engineering, business, and strategy, etc. The opportunities for systematic tool developments identified by the authors include but not limited to the following:

- More systematic resource identification algorithm/processes for various Useless-to-Useful and Harm-to-Help situations.
- Systematic identification of new perspectives for problem understanding and solving. That is: How to find all or some other Space-Time-Domain-Interface to perceive the problem.
- Systematic approach to identify a new system to transfer the issue of the current problem to.
- Systematic approach to identify new paradigms to shift to in order to solve problem innovatively.
- How to convert from a Model of Solution to Specific Solutions effectively and hopefully comprehensively?

To the best of author's knowledge, the above-mentioned tools are not available on literature yet except for (Sheu & Hong, 2017) where systematic methods and practical examples were provided to achieve U2U resource utilizations. Items on the list thus provides good directions for new tools development toward realization of the powerful TRIZ philosophies.

4. Summary and conclusions

Most of TRIZ researches focus on developing tools for problem solving and most of application practitioners focus on using various TRIZ tools to solve problems. This paper particularly address the fundamental philosophies which enable TRIZ tools to work powerfully and in wide range of applications. These philosophies facilitate TRIZ tools to break psychological inertia which is a main source for breakthrough innovative ideas. The summarized 7 pillars of philosophies are: 1) Ideality, 2) Resources, 3) Functionality-Value, 4) Contradiction, 5) Space-Time-Domain-Interface, 6) System Transfer, and 7) System Transition. By identifying the fundamental philosophies behind TRIZ working principles, this study also identifies some directions for new TRIZ tool development to bridge the working philosophies and the real-world applications by developing practical hands-on tools as indicated in the previous sections.

CRediT authorship contribution statement

Dongliang Daniel Sheu: Conceptualization, Investigation, Methodology, Writing - original draft, Supervision. **Ming-Chuan Chiu:** Resources, Data curation, Writing - review & editing, Project administration, Funding acquisition. **Dimitri Cayard:** Investigation, Writing - original draft.

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Appendix A. Supplementary data

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